

Carrier 38MURA Service Manual (2019)

Model 38MURA | 2019 | Service

SM-38MURA-2019

Prototype sample document for the clickable demo. Not an official publication.

Rev	Date	Description
1.0	2019-05	Initial release for the 2019 model year.
1.1	2019-09	Updated CN2 pinout and fault-code table.
1.2	2020-02	Added inverter diagnostics and torque values.

> For professional use only. Service work must be performed by qualified personnel.

Contents

Safety and Tools	3	Sensor Diagnostics	16
System Overview	4	Overcurrent Protection	17
Call for Fan Sequence	5	Control Board Replacement	18
Error Code E5	6	Post-Replacement Checks	19
E5 Measured Values	7	Maintenance Schedule	20
Blower Motor	8	Troubleshooting Quick Reference	21
Blower Motor Testing	9	Error Code Appendix	22
Control Board Overview	10	Specifications	23
Board LED Codes	11	Electrical Characteristics	24
LED Code Reference	12	Parts Replacement Reference	25
Inverter Diagnostics	13	Glossary	26
Wiring Diagram	14	Revision History	27
Wiring Legend	15	Warranty and Notes	28

> Page numbers refer to the printed page footer. Cross-references use section numbers.

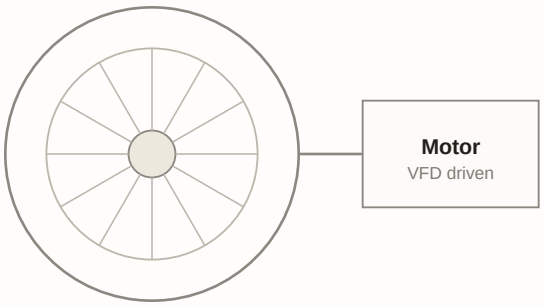
3. Safety and Tools

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Safety and Tools reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

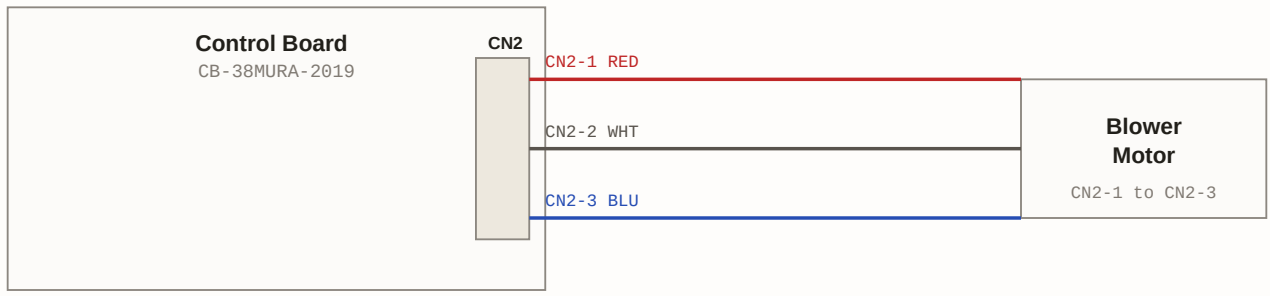
4. System Overview

Connector CN2 carries the low-voltage control signal between the board and the blower motor. A loose or corroded pin here is one of the most common causes of intermittent operation.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

System Overview reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

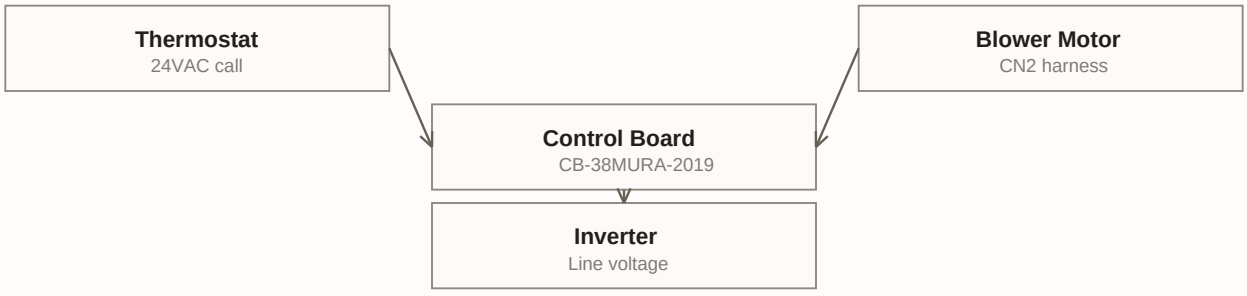
5. Call for Fan Sequence

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Call for Fan Sequence reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

Error Code E5: AC Overcurrent

Section 5.2: Fault Codes & Diagnostics

E5 indicates an AC overcurrent condition detected at the inverter output.

E5 is logged when the board senses current above the upper limit at the inverter output. In most cases the condition is caused by a missing or intermittent control signal rather than a failed motor. Verify the blower control signal before condemning any component.

Common causes

- Lost or intermittent 24VAC control signal at CN2.
- Loose or corroded CN2 pins.
- Blower motor overload or seized bearings.
- Failed control board output stage.
- Incorrect field wiring at the thermostat.

Step	Check	Expected
1	Line voltage at unit	208-230 VAC
2	24VAC at CN2-1	24 VAC +/-10%
3	Motor spin by hand	Free rotation
4	Board LED code	See Section 6.3

> Always confirm 0VAC and 0VDC at the board before making resistance measurements. See Appendix D for the CN2 wiring diagram.

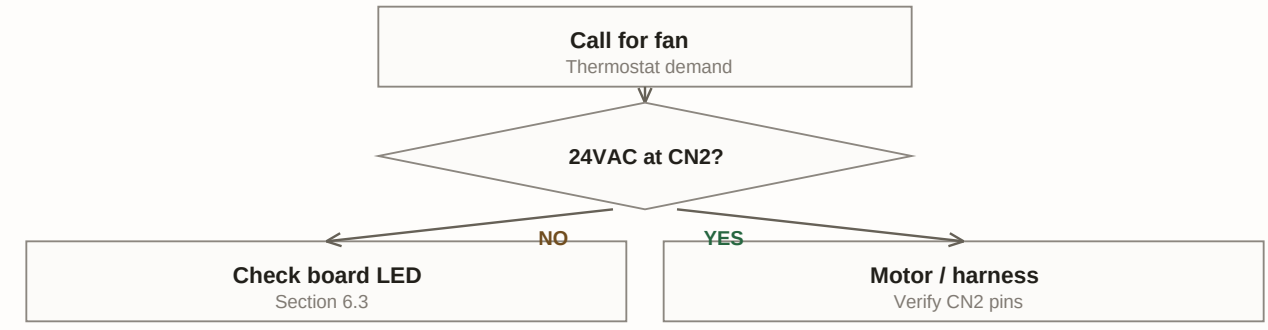
7. E5 Measured Values

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



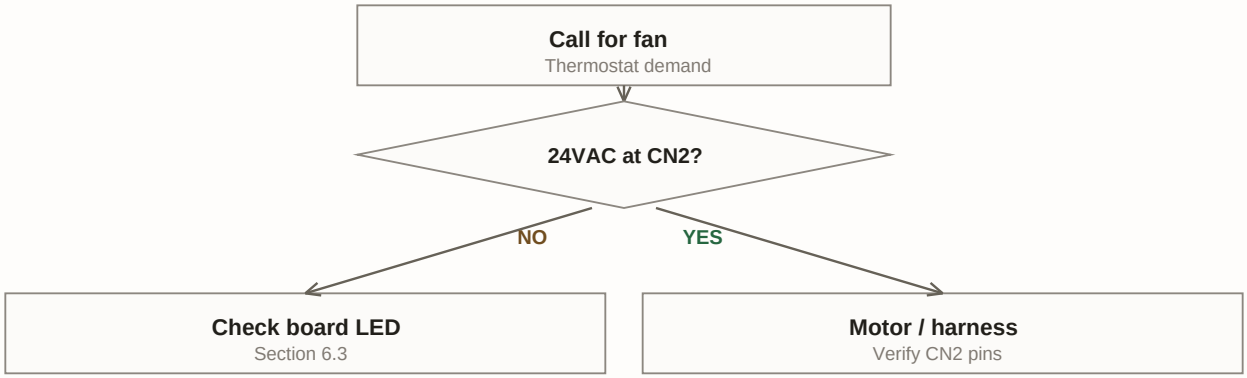
E5 Measured Values reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

Blower Motor: No Spin

Section 5.4: Indoor Blower Diagnostics

A blower that will not spin during a call for fan is most often caused by a missing 24VAC control signal at CN2. Work through the flow below before removing the motor.



Blower no-spin diagnostic flow

Measurement	Point	Range
Control signal	CN2-1 to CN2-2	24 VAC
Motor resistance	CN2-3 to CN2-2	see chart
Tach feedback	CN2-3	pulsed DC
Line voltage	L1 to L2	208-230 VAC

> If the motor spins freely but receives no control signal, the fault is upstream at the control board.

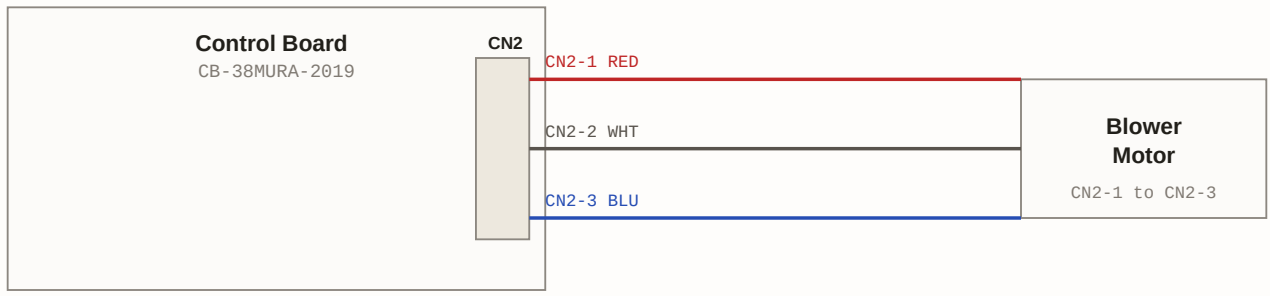
9. Blower Motor Testing

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

Blower Motor Testing reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

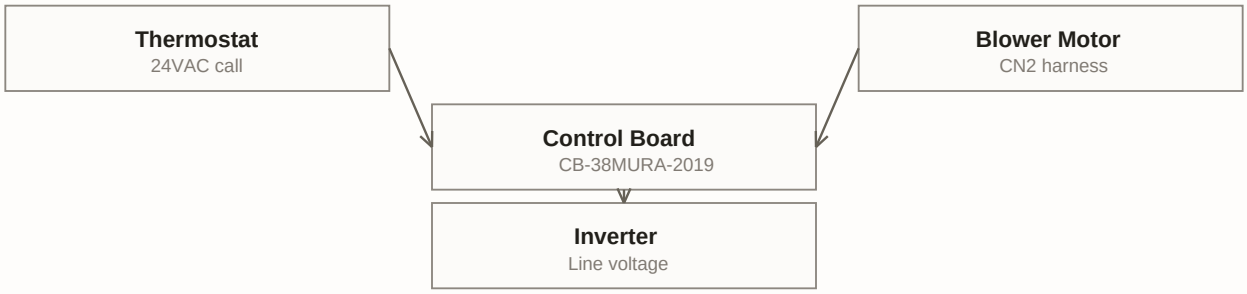
10. Control Board Overview

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Control Board Overview reference

> If a code returns after clearing, inspect the harness before replacing the board.

Control Board Fault: 3-Flash Code

Section 6.3: Board LED Diagnostics

3 flashes: internal control-board fault, replace the board.

The control board LED reports a fault code at startup. Count the flashes after the board completes its self-test. Confirm the harness and the low-voltage transformer before ordering a replacement.

Flashes	Meaning	Action
1	Line-voltage fault	Check supply and fuses
2	Low-voltage fault	Check transformer
3	Board fault	Replace board
4	Sensor fault	Check sensor wiring
5	Communication	Check harness

Replacement decision

1. Confirm 24VAC at the transformer secondary.
2. Confirm the harness is fully seated at CN1 and CN2.
3. Re-check the flash count with the unit calling for fan.
4. If the code returns, replace the control board CB-38MURA-2019.

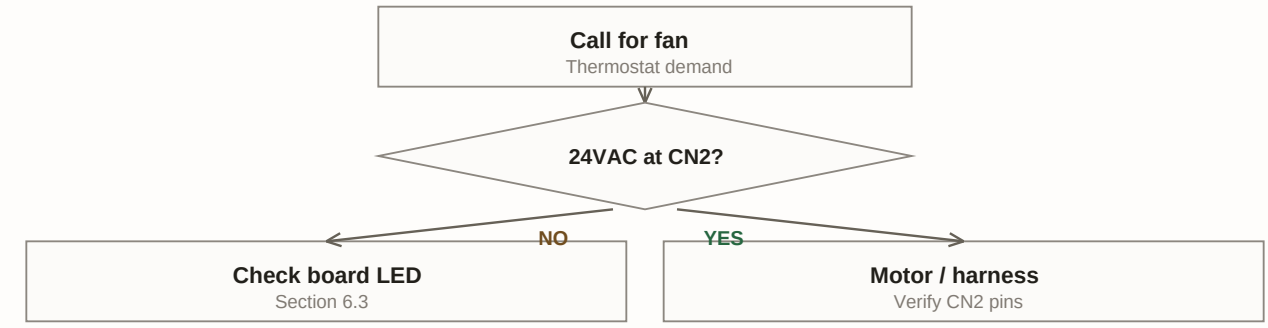
12. LED Code Reference

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



LED Code Reference reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

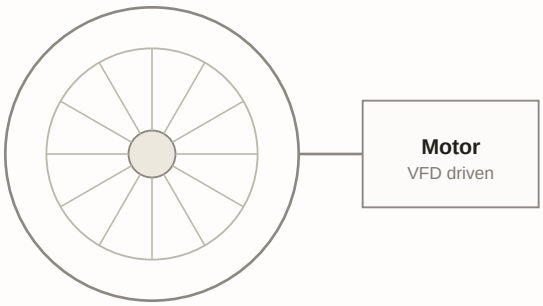
13. Inverter Diagnostics

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



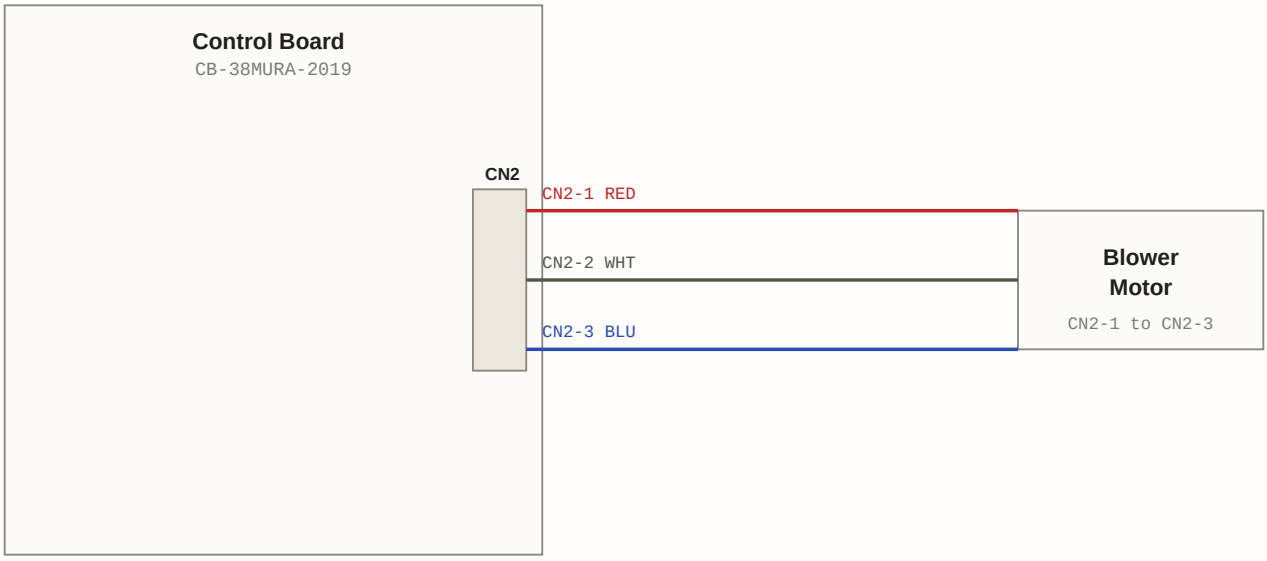
Indoor blower wheel and drive motor
Inverter Diagnostics reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

Wiring Diagram: Control Board CN2

Appendix D: Schematics

The low-voltage control signal leaves the board at connector CN2 and runs to the blower motor harness. Verify each pin against the table before probing the board.



24VAC control signal from board to blower motor

CN2 control-signal wiring

Pin	Wire	Signal
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback

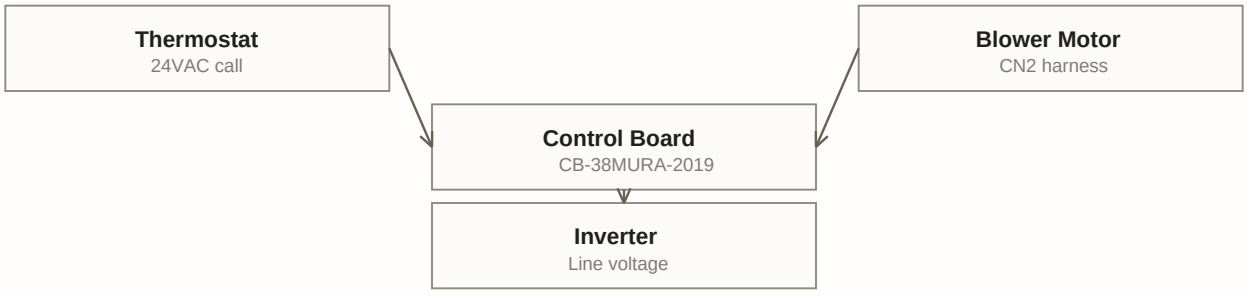
15. Wiring Legend

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Wiring Legend reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

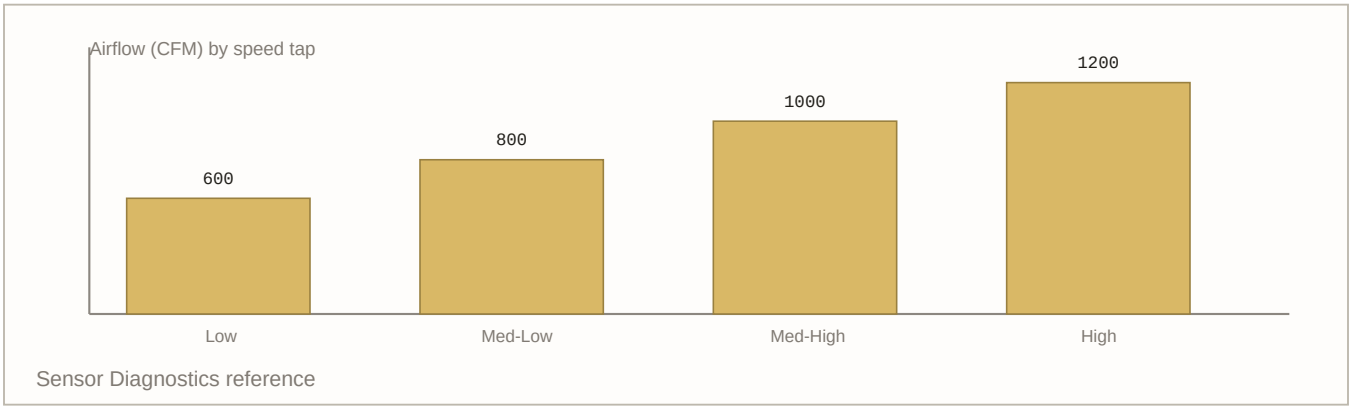
16. Sensor Diagnostics

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Sensor Diagnostics reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

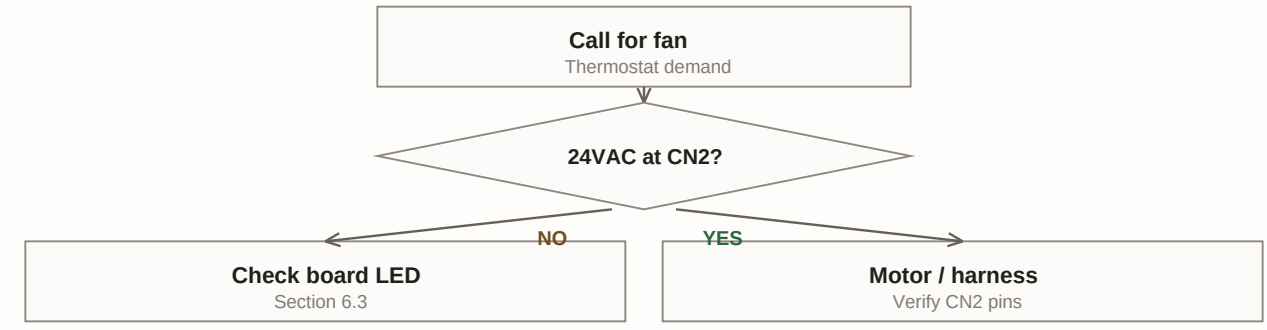
17. Overcurrent Protection

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Overcurrent Protection reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

Control Board Replacement

Section 7.1: Control Board (CB-38MURA-2019)

Use the procedure below to replace the control board. Transfer the configuration jumper from the old board so field settings are preserved.

1. Disconnect power at the service switch and confirm 0VAC at the board.
2. Remove the control-box cover and photograph the harness routing.
3. Disconnect CN1, CN2, and the thermostat terminal block.
4. Remove the two mounting screws and lift the board out.
5. Transfer the configuration jumper to the replacement board, then reverse the steps.

Item	Tool	Torque
Mounting screws	1/4 in driver	2.5 N-m
Terminal screws	Torque driver	0.8 N-m
Ground screw	1/4 in driver	2.5 N-m

> Run a full startup sequence after replacement and confirm the LED shows a steady state with no fault code.

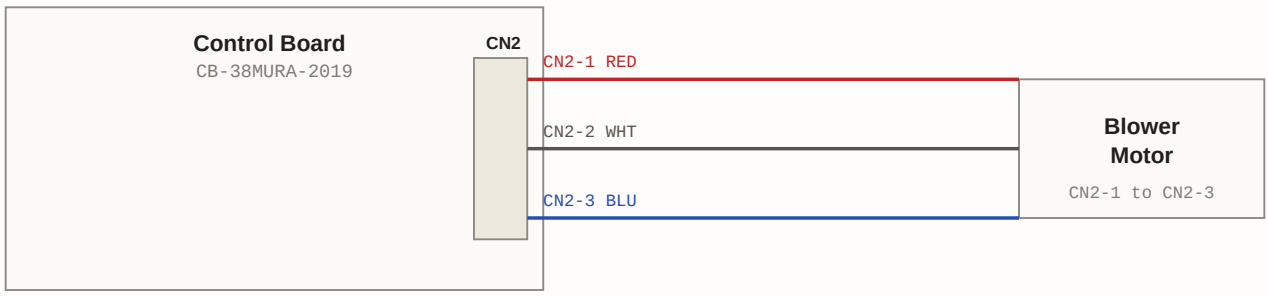
19. Post-Replacement Checks

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

Post-Replacement Checks reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

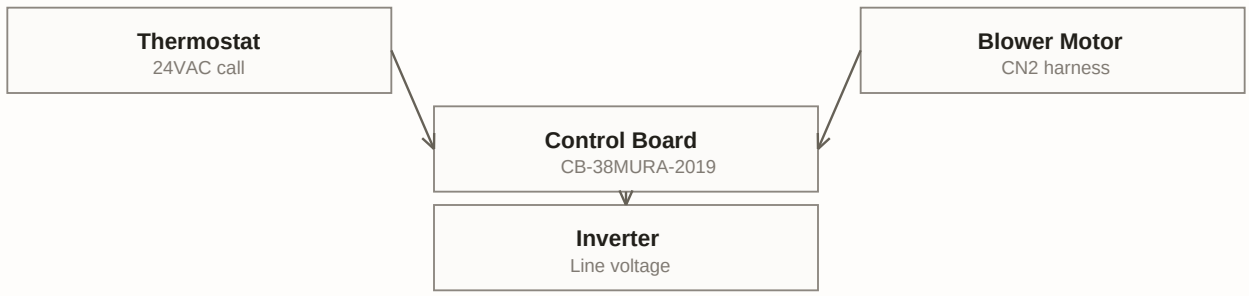
20. Maintenance Schedule

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Maintenance Schedule reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

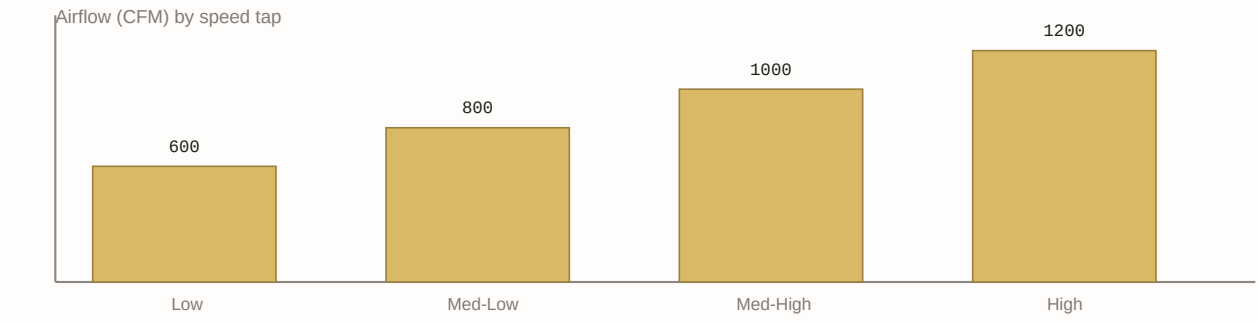
21. Troubleshooting Quick Reference

The 38MURA uses a variable-speed inverter to match capacity to load. Because the inverter regulates motor speed, most faults present as a control-signal problem rather than a mechanical failure.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Troubleshooting Quick Reference reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

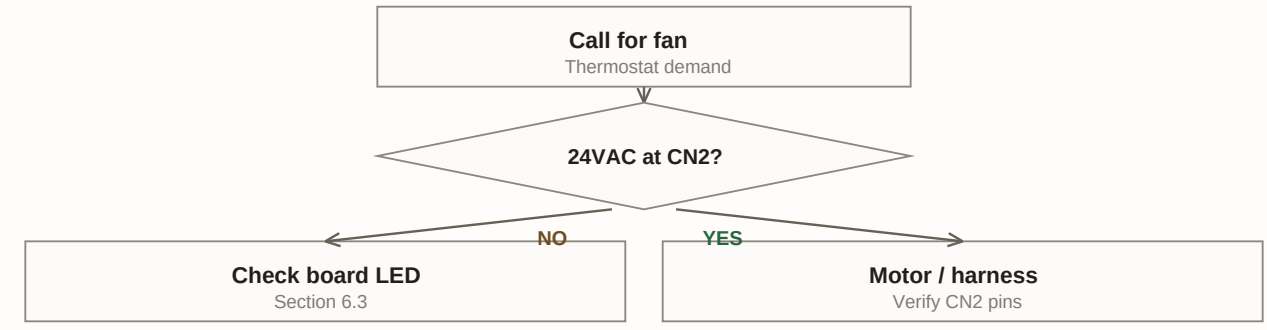
22. Error Code Appendix

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Error Code Appendix reference

> If a code returns after clearing, inspect the harness before replacing the board.

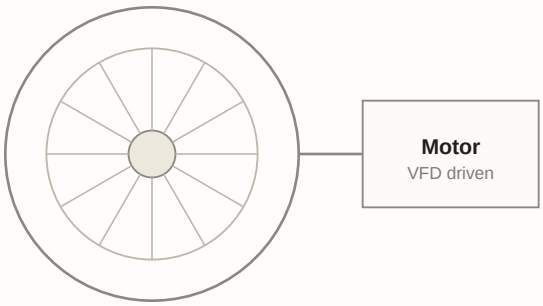
23. Specifications

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Specifications reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

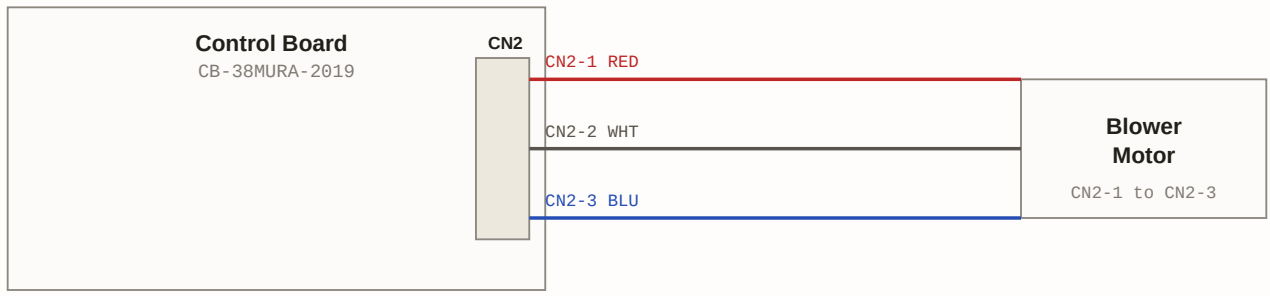
24. Electrical Characteristics

Connector CN2 carries the low-voltage control signal between the board and the blower motor. A loose or corroded pin here is one of the most common causes of intermittent operation.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Speed Tap	CFM	Static
Low	600	0.20 in wc
Med-Low	800	0.30 in wc
Med-High	1000	0.40 in wc
High	1200	0.50 in wc



24VAC control signal from board to blower motor

Electrical Characteristics reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.

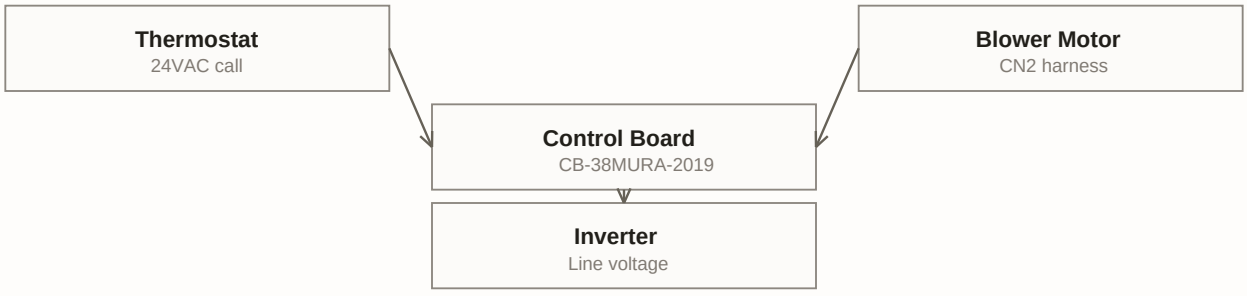
25. Parts Replacement Reference

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
DC bus	310 VDC	Discharged before service
CN2 signal	24 VAC	Call for fan
Ambient	see chart	Record at test



Parts Replacement Reference reference

> Record the outdoor ambient and the suction line temperature together for an accurate comparison.

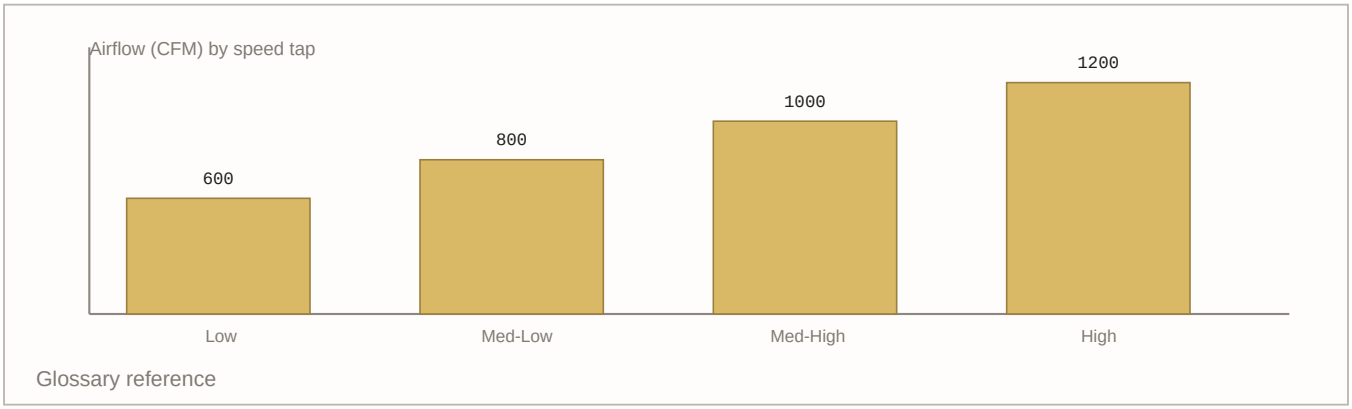
26. Glossary

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
CN2-1	RED	24VAC supply
CN2-2	WHT	Common return
CN2-3	BLU	Tach feedback
CN1-1	BLK	Earth ground
CN1-2	YEL	Thermostat Y



Glossary reference

> If a code returns after clearing, inspect the harness before replacing the board.

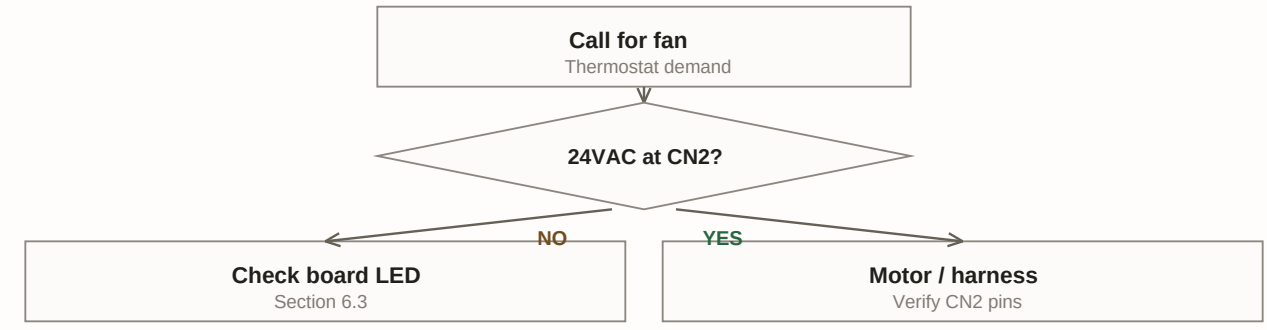
27. Revision History

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E4	Locked rotor	Check compressor
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness



Revision History reference

> Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

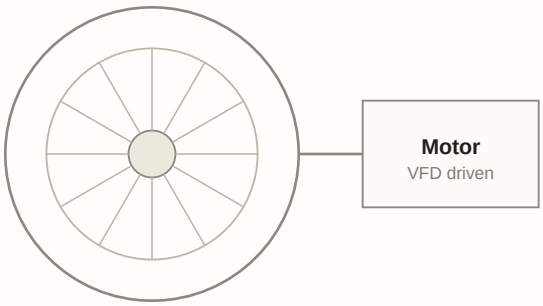
28. Warranty and Notes

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the board terminals over time.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Step	Tool	Spec
Mounting screws	1/4 in driver	2.5 N-m
Terminals	Torque driver	0.8 N-m
Line set	Flare wrench	per chart
Filter	Hand	snug
Panel	1/4 in driver	1.5 N-m



Indoor blower wheel and drive motor
Warranty and Notes reference

> Always confirm 0VAC and 0VDC at the board before making resistance measurements.